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**USE OF AUGMENTED REALITY WITH
GAMIFICATION ELEMENTS FOR MARKETING
ON SOCIAL NETWORKS**

BACHELOR THESIS

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Declaration

I declare that I worked on the bachelor thesis "Use of augmented reality with gamification elements for marketing on social networks" independently using the sources and literature mentioned in the thesis.

Prague, May 10, 2021

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Abstract

This bachelor thesis' topic is assessing the current state of AR. It focuses on the historical development of this technology and its current use for marketing purposes on social networks. The thesis contains an analysis of the program for AR effects creation - Spark AR by Facebook, the process of creation of interactive AR effect with gamification elements that can be used in the commercial sphere, and a description of the implementation of the effect on social media and its evaluation.

Keywords

Augmented reality, social networks, Spark AR, AR effects

Abstrakt

Tématem této bakalářské práce je rozbor současného stavu rozšířené reality. Práce se zaměřuje na historický vývoj této technologie a její současné využití pro marketingové účely na sociálních sítích. Práce obsahuje analýzu programu pro tvorbu AR filtrů - Spark AR od společnosti Facebook, proces tvorby AR filtru s prvky gamifikace použitelného pro komerční účely, popis implementace filtru na sociální síť a jeho zhodnocení.

Klíčová slova

Rozšířená realita, sociální síť, Spark AR, AR filtry

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Introduction

Marketing managers are constantly trying to find new ways to get as close as possible to customers and sell them a certain product. In today's market, where the customer has a choice of an inexhaustible number of brands, it is crucial for the brand to differentiate itself in some way and attract the customer's attention. It is therefore crucial for the marketer to constantly monitor current trends and use new methods and technologies for promotion.

Another current trend in this industry is the use of augmented reality. Augmented reality allows the customer to interact more closely with the product and therefore there is a greater chance that with this immersive experience, the customer will create a certain memory of the product. As with any other promotion, even when using augmented reality for marketing purposes, it is important to consider how to use it to have the right impact on the target group and meet the brand's expectations. Behind such promotion is a certain process which this thesis deals with.

The augmented reality experience can be transmitted using various devices. Each of these devices works on slightly different principles and brings a slightly different experience. As a relatively young technology, it is constantly evolving and subject to further research. In some industries, this technology has caught on quickly and even makes work easier for people, while its adaptation to everyday life is worse and slower.

Currently, the general public most often encounters augmented reality on social networks. Large social networks even make available free software to the public so that ordinary people could create augmented reality for their platforms. This fact plays into the hands of brands being promoted on these platforms. This is still a substantially new way of promotion, and specifically on the Instagram social network, with usage of AR filters a high organic engagement can be achieved - which is really rare nowadays.

In the first few parts, this thesis deals with the development and definition of augmented reality and subsequently with current technologies and devices used for augmented reality and their use in practice. Subsequently, it focuses in more detail on the use of augmented reality on social networks. The last part deals with the process of creating a promotional interactive AR filter to achieve organic engagement on the social network Instagram.

1 Definition of realities

Augmented reality is located in the middle of the reality-virtuality continuum defined by Paul Milgram (1994) in his article on A Taxonomy of Mixed Reality Visual Displays that he wrote in Kyoto at the Institute of Electronics, Information and Communication Engineers during his research on mixed reality. The reality-virtuality continuum describes the journey from the physical world to the fully immersive experience of virtual reality.

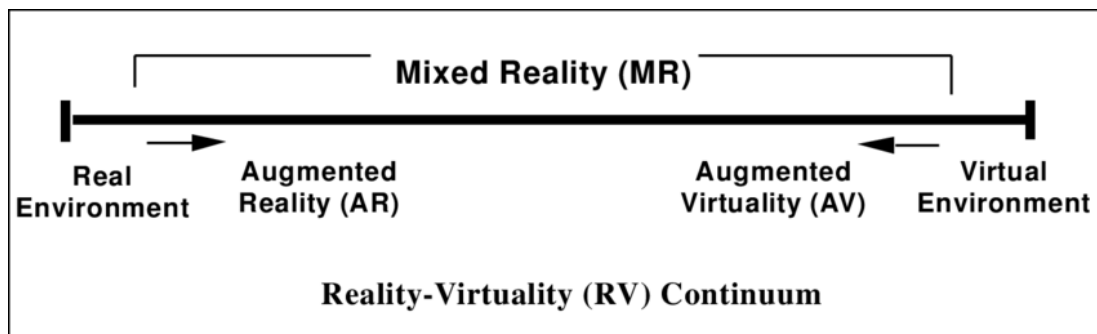


Figure 1 — Representation of a RV Continuum

Real Environment

The schema begins with the real world, which is not extended by any digital elements. It is simply an existing environment that is captured with human senses.

Augmented Reality

By using a smartphone, tablet, or smart glasses you can get into an augmented reality environment. Augmented reality is a real environment enriched with digital elements such as pictures, videos, information, or objects (Dalton, 2021).

Augmented Virtuality

In an environment of augmented virtuality, it is worked mainly with 3D objects that can be anchored to a certain place in the real environment. Augmented virtuality is often associated with augmented reality because both provide a similar experience. The main difference is that for full-fledged augmented virtuality, the device used for the experience must be able to understand the real environment and recognize where objects are located (Dalton, 2021).

Virtual Environment

Users can immerse themselves in the world of virtual reality with the help of a VR headset or surrounding display. There is nothing left of the real environment in this case. Virtual reality can be artificially created by a computer or digitized by a real environment (Dalton, 2021).

2 History of Augmented Reality

To understand the current status and possibilities of augmented reality, it is necessary to focus on its development and history. The first signs of augmented reality can be dated back to 1967. That year, the Harvard professor and computer scientist Ivan Sutherland and his students designed the first VR / AR headset called "The Sword of Damocles" (Greengard, 2019). The user experienced computer-generated graphics that enhanced their sensory perception of the world. The headset was mounted to the ceiling due to its weight. The moment the user put on the headset, it started working (Poetker, 2019). The user could still see the real environment around them, but it was supplemented by graphic elements - with this, it can be talked about the first augmented reality experience.



Figure 2 – VR / AR headset; Figure 3 – AR experience

Another milestone in the development of augmented reality is Videoplace, which was created by the computer researcher and artist Myron Krueger from the University of Connecticut (Poetker, 2019). Krueger has been working on human-computer interaction since the 1960s. He describes all his projects in his book *Artificial Reality II*. His motivation was a desire to solve real practical problems, but he also admits in the book that apart from solving problems, he was driven by wanting to find aesthetic solutions. His most influential project Videoplace is based on his previous projects Glowflow and Metaplay and he mentions in *Artificial Reality II* that its origin was a coincidence. Videoplace was based on the principle of projection and camera technology. The user was surrounded in the room by cameras that captured the user's movement and a video-projection screen on which the user was displayed and could interact with objects on the screen. The user could paint, move objects, or type on a digital keyboard (Krueger, 1991).



Figure 4 – Videoplace

In 1990, the term "augmented reality" first appeared. This term was defined by Boeing researcher Tom Caudell in an effort to make work more efficient for technicians and operators that were assembling a 777 jet. The technicians had to constantly flip through lengthy instructional sheets, and this unnecessarily lengthened the process. Caudell and David Mizell (his colleague) had an idea. They wanted to create a see-through display for technicians and operators that could guide them through the process by superimposing lines for where the wires should go on top of the board, and all documentation was to be stored inside. The fact that the 777 jet was the first model to have all its components fully digitized before it was physically assembled also played into their hands (Metz, 2014). It was another attempt at smart AR glasses.

Two years later, the first functional AR system appeared in the USA - Virtual Fixtures. The system was designed as a training instrument for the US Air Force. The purpose was to provide pilots with the possibility of training in safer flying practices (Rosenberg, 1992). The creator of Virtual Fixtures, Louis Rosenberg, not only came up with a functional system, but due to the fact that the system worked with both graphic elements in glasses and physical objects Rosenberg can be considered as a pioneer of augmented virtuality or mixed reality.

During the last decade of the 20th century, new signs of augmented reality in culture and sports began to appear. It is definitely worth mentioning the yellow line which you can see on television while watching American football, which represents the distance the offense needs to go to get a first down. It first appeared on television screens in 1998 during a live NFL broadcast (Poetker, 2019). Similar elements are nowadays used in various live sports broadcasts.

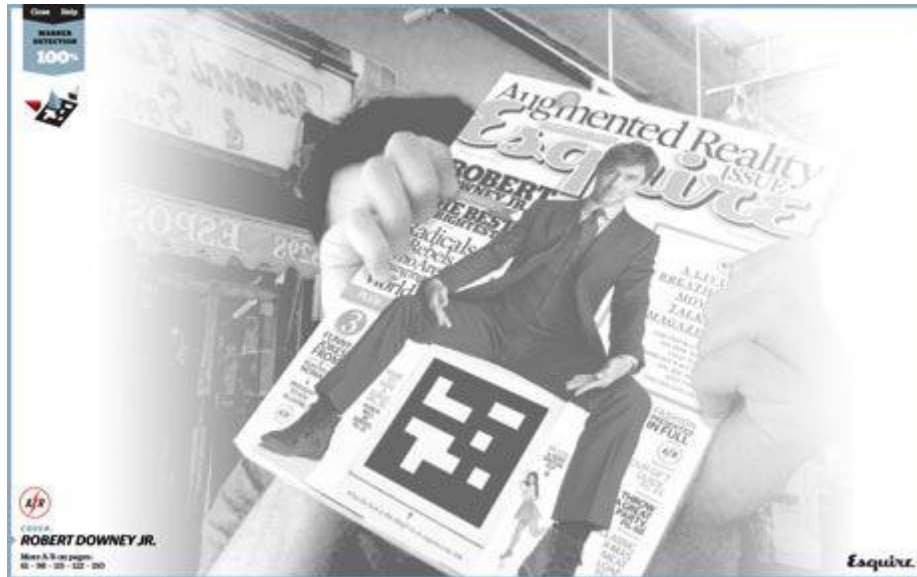


Figure 6 – Esquire cover with AR experience

The same year Microsoft introduced a revolution in the gaming industry - Kinect. At the conference in June 2009, Don Mattrick was joined on stage by film director Steven Spielberg, who presented the potential of Kinect (Hester, 2020). This new technology, which works with the Xbox game console, can draw the player more into the game as the player uses only his body to control the game and does not need any controller.

AR technology is also used by Volkswagen to facilitate work. In 2013, they came up with their MARTA application (Mobile Augmented Reality Technical Assistance), which streamlines and speeds up the car repair process (Poetker, 2019). MARTA guides the technicians step by step during the repair and there is no need for a lengthy reading of the manuals. The only thing the technicians need is a tablet. The technician points the tablet at the component that is about to be repaired and the application guides the technician in real time using infographics (Lee, 2013).



Figure 7 – Visualization of the MARTA application

The Google Corporation, which in 2012 introduced its first AR glasses concept, is not left behind either. In April 2012, they released the video "Project Glass: One Day," which shows what a normal New York citizen's day wearing smart glasses would look like (Google, 2012). With the help of Google smart glasses, user can make phone calls, use navigation, take photos and share them with others, track the locations of friends, or, for example, record reminders in the calendar - all with voice control. A few months later, Google showed the first demo product, Google Glass, at its conference (Poetker, 2019). Sergey Brin communicated with parachutists that were about to jump out of a plane while he was on stage. Brin and the parachutists were all wearing Google smart glasses. The entire video call was screened live at the conference with the audience (Google Developers, 2012). The first Google glasses were launched in 2013 for \$ 1,500. Soon, a discussion about privacy was sparked around the Google glasses, and some businesses and restaurants even banned wearing them on their premises (Betters, 2015).

In 2016 developers could look forward to mixed reality smart glasses from Microsoft called HoloLens. HoloLens glasses, unlike Google Glass, can be controlled by gestures, track head movement, track the user's position in space and warn the user if there is any obstacle in their way. On the other hand, HoloLens glasses were twice as expensive as Google Glass and more than ten times heavier (Versus, 2021). In the first promo video for them from 2015, Microsoft presents what HoloLens glasses can be used for. In the video Microsoft mentions not only communication between people but also a better gaming experience, a potential for education, smart home improvements, and product designing as advantages that the glasses can provide (Tech Discussion, 2015).

In autumn 2015, Snap Inc. introduced a new feature on its social media platform Snapchat – Lens (Newton, 2015). This feature allows users to add real-time effects into their photos by using face detection technology.

Augmented reality in smartphones became a hit among the public not only due to the face-filters on Snapchat but also thanks to the mobile game Pokemon GO from Niantic in 2016 (Poetker, 2019). During its first year, the application had 500 million downloads worldwide (Iqbal, 2021).

Ikea also provides an immersive experience to the public. This time it's not about fun, but about using augmented reality to buy goods. Ikea released the "PLACE" application, with which the user can try out furniture at home without any unnecessary re-measurement and with just a few clicks (Dasey, 2017). The user first scans the environment and chooses a piece of furniture from the catalog. Then the user virtually places the object in the environment and by moving the phone, the user can properly view the object in the room - whether it fits in terms of color and size. The application was initially supported only in iPhones and iPads (Ikea, 2017).

3 Current technology designed for AR

Technology has shifted significantly since I. Sutherland built the first smart glasses. Currently there exist several types of augmented reality devices that work on different principles. Each type is able to give slightly different experiences. These devices can be divided into the following categories: heads up displays (HUD), holographics displays, smart glasses, and handhelds (Kore, 2018).

3.1 Heads up displays (HUD)

HUD technology projects information directly onto the user's field of vision. The advantage of heads-up displays is the fact that this technology takes into account the position of the user's head and his view (Kore, 2018). A heads up display works on the basis of these three basic components - the projection screen (glass), the projector, and the computer. The projector and computer can be summarized as an "image-generating unit". This unit has the function of producing the respective information and transmits it visually as an image, video, or 3D animation onto the relevant projection surface. The projection screen must meet two main criteria - it must be transparent, but at the same time, it must be projectable (Mangold, 2021).



Figure 8 – representation of HUD in car

Heads up displays can be imagined in several forms. The most used HUDs are those on the windshields of cars or planes (Roberson, 2021). Navigation, the weather, and

information about the surroundings of the vehicle are the most common types of information projected on the glass. The purpose of the HUD on the windshield is to enable pilots to look forward outside rather than looking down inside the cockpit. This keeps their attention on the “road.” HUDs in cars and airplanes usually use a lower quality - mostly FullHD, as the aim of the screening is mainly to convey relevant information (Mangold, 2021).

High-quality HUDs or so called Deep Frames are used, for example, in museums, showrooms, or exhibitions (Mangold, 2021). These HUDs can be thought of as transparent mirrors or windows. The immersive experience happens via a projection on the glass and creates the illusion that the projected objects are actually behind the glass. Such a result is achieved because high-quality HUD’s work with 4K UHD quality and also because this technology works with the viewer's position in space and the angle from which the viewer watches the screen (Hardawar, 2018).



Figure 9 – Deep Frame used for promotion in a conference

In aviation, this technology has really found its justification and is taken for granted. In addition to the HUDs on the windshields of aircraft, helmet-mounted displays are also used in aviation. The user wears a helmet in which both the image-generating unit and the projection glass are built. The moment the user puts on his helmet, he sees the surroundings around him, but at the same time information, pictures, etc. are projected directly in front of his field of vision. The helmet tracks the user's head movement, so the projected content can remain relevant to the environment (Mangold, 2021).

3.2 Holographic near-eye displays

Another technology that is able to provide us with an immersive experience is holographic near-eye displays. An example of such technology is the HoloLens glasses by Microsoft. These glasses work according to the principle of light diffraction techniques (Kore, 2018).

3.2.1 Smart Glasses

As augmented reality began to spread to the general public, it made sense to devise an augmented reality device that could be easily controlled by anyone. Smart glasses' users simply put them on their head. Subsequently, computer-generated or digital information such as 3D objects, videos, images, or animations begin to appear in the user's field of vision (Software Testing Help, 2021). Most smart glasses do not work on their own and must be connected to other devices - such as a computer, a smartphone, or some other device. Bluetooth or Wi-Fi is most often used to connect to these devices. There are currently a lot of types and brands of smart glasses. The most common division of smart glasses is into optical see-through and video see-through glasses (Kore, 2018).

Optical see-through

Smart glasses, which work according to the principle of optical see-through, project information to the user on a transparent material - usually glass. Thus, the user sees the real environment through the glass while this is supplemented by various graphic elements. This information is projected on the glasses using holographic technology or other systems that enable graphical overlay in the real environment (Kore, 2018).

HoloLens

As was mentioned above, this category includes the smart glasses made by Microsoft called HoloLens. These glasses work according to the principle of holographic processing. The first HoloLens glasses, which were announced in 2009, were replaced by a more current version - HoloLens 2. These glasses can be purchased since 2019. In ten years, the HoloLens glasses have undergone significant improvements (Versus, 2021).

A new feature in HL2 is built-in eye-tracking, which serves for more natural interaction. Eye sensors monitor the movement of the eye and adjust the position of objects on the glasses based on the user's direction of view. It includes iris recognition, which can recognize the glasses user by using their iris. The user can use this function as an alternative to passwords (Software Testing Help, 2021).

HoloLens works without controllers and purely on the basis of voice control and hand tracking. Now the user can use both hands unlike in the previous model, where it was possible to interact with one hand only. At the same time, thanks to hand tracking improvement, users can use gestures for object control. The hand-sensing sensor can detect up to 25 points of articulation on the wrist and fingers. Gestures such as Press, Grab, Direct manipulation, Touch interaction or Scroll with wave have been added to the existing finger taps and palm movements. These gestures allow more sophisticated movement

and manipulation with objects. Users can rotate, enlarge, or reduce objects, or place objects in hands, all with the help of gestures (Microsoft, 2020).

It is definitely worth mentioning the change of the field of view, which has almost doubled since the first HoloLens edition - to 52 °. In HL1, in some cases, the user had to turn his head sideways to see the entire object. The display resolution has also been increased from 1280x720 px (per eye) to 2048x1080 px (per eye) (Versus, 2021).

Google Glasses

Despite the epic show with parachutists mentioned above, the first Google Glass model did not reap such success among the public - mainly due to privacy issues. Therefore, in 2017, the company decided to focus on smart glasses for companies and businesses, where this solution makes more sense. Google currently has 4 types of Google glasses in its portfolio - Explorer Edition and Explorer Edition 2 (focused on consumers), and Enterprise Edition and Enterprise Edition 2 (focused on businesses) (Václavík, 2019).

Both types of Explorer glasses - Explorer Edition 1 and Explorer Edition 2, have on the right hand side a trackpad, a glass prism right above the eye and a projector that projects information into the glass prism. The user operates the glasses by voice or by swipe gestures on the track pad. At the same time, the information is not projected into the entire field of view, but only on the upper right part, where the prism is located. These two editions are mainly used for navigation and video recording, or when they are connected to a phone, the user can ask questions or send messages (Versus, 2021).

Enterprise Edition 1 and Enterprise Edition 2 for businesses are not significantly different in their appearance compared to the Explorer Editions. The size of the projected screen remains the same (640x360 px) and the glasses are also controlled by voice and trackpad. However, Google is promising better performance and endurance in this case, which should be achieved by the Enterprise glasses' increased battery capacity. The Enterprise Editions differ from the Explorer Editions mainly in terms of their chipset. Edition 1 has an Intel Atom, and Qualcomm Snapdragon XR1 was chosen for Edition 2. The storage has also been improved - from 16GB to 32GB - and in Edition 2 we can also find an improved camera with 8 Mpx (Versus, 2021).

Vuzix

The smart glasses by Vuzix work according to the same principle as Google Glass. Vuzix currently offers 4 different types of AR glasses - M4000, M400, M300XL and Blade Upgraded. Their main target group are businesses and therefore their designs focus mainly on technical specifications rather than appearance. For example, their latest model, the M4000 model, is able to record video in 4K quality with 30fps. Consumers are partially targeted with the Blade Upgraded type, which has a modern design resembling sunglasses and the projector is built into the rims (Vuzix, 2021). Unlike in the case of the Google Explorer Editions, the image is projected directly on the glasses in this type.

Snap Spectacles

Augmented reality glasses are also being developed by Snap Inc. - the company that owns the Snapchat social network. Thanks to this social network and its face filters, interest in augmented reality has increased. The company wanted to take the immersive experience of their filters a bit further and created its first smart glasses - Snap Spectacles - in 2016. Users connect these glasses to the Snapchat application and by using the button on the frames of the glasses, they can record ten-second videos (Low, 2019). The main advantage of the Snap Spectacles is that users can apply Snapchat filters to videos and see that right in the glasses. After a video is recorded, it is saved in the Snapchat application and the user can share it with friends using their phone. The latest version of the glasses - Snap Spectacles 3 - was released in 2019. This version offers a dual-camera setup and allows the user to capture not only videos and photos but also photos in 3D. Another advantage of the dual-camera setup is the fact that its smart AR filters respond to the environment. In previous models, AR filters could be applied only as a flat overlay; however, in this new model, it is possible to work functionally with 3D objects. (Spectacles, 2021)



Figure 10 – Snap Spectacles 3

Many companies worldwide are currently working on the development of AR glasses and they can be bought in various price ranges. Glasses that work on the same principle as HoloLens, are worth a mention. Such glasses are produced by HoloLens' competitor Magic Leap. Today's other smart glasses are the following brands: Epson Moverio, Echo Frames, Focals by North, Thinkreality by Lenovo, and Huawei Gentle Monster Eyewear (Kore, 2018).

It is speculated that next year, in 2022, the Apple company will release its own smart glasses. The glasses should be casual and meet Google's initial expectations. Facebook is also reportedly working on AR glasses (Global Data, 2021).

Video see-through

The second type of AR glasses is video see-through glasses. This type of glasses first captures reality using cameras built into the glasses and projects the videos or photos to the user on the screen with computer-generated elements. This category includes, for example, the HTC Vive VR and Valve Index headsets. These headsets are primarily intended for virtual reality but also enable an augmented reality function (Kore, 2018).

Handheld AR

In most cases, the consumer society will experience augmented reality through smartphones or tablets. The user can access the immersive experience using an AR based application or a web browser with elements of AR, or nowadays via social networks such as Instagram, Facebook or Snapchat.

4 Social networks using AR

In most cases the general public experiences augmented reality through social networks. The first encounter with augmented reality on social networks was on Snapchat in 2015 (Newton, 2015). Since then, more social networks have started to implement this feature into their platforms. Currently, augmented reality is found on almost all large social platforms such as Facebook, Instagram or TikTok.

4.1 Snapchat

Snapchat is a mobile application that works as a chatting platform. Users communicate with each other via chat and mostly using "snaps" - i.e. photos. The user can take a photo, select a photo from the phone's gallery, or record a video. The user can add text, stickers or AR lenses to the photo or video and send it as a private message to one or more users. Another option is to share the photo publicly, and all the user's friends can see it for twenty four hours (Tillman, 2021).

There are lenses and 3D world lenses in the application. Lenses are augmented-reality-based effects that are applied to the user's face, feet or hands. The function of lenses is to diversify a photo, beautify/distort a face or interact with AR elements for fun. 3D world lenses are also augmented-reality-based effects. These kinds of effects interact with the real environment that the user is surrounded by. To activate 3D world lenses, the rear camera of the phone must be used for the photography or video. The user can, for example, place various 3D objects into the real environment (Tillman, 2021).

4.1.1 Lens Studio

The feature of lenses and 3D world lenses became very popular among Snapchat users. Regarding Snap Inc. internal data from Q4 2019, on average over 75% of the Snapchat community engages with augmented reality to communicate, play, and learn every day. For this reason, Snapchat decided to make the creation of AR effects available to the public and released its own augmented reality software - Lens Studio. With Lens Studio by Snap, anyone can create their own effects and upload them to Snapchat, where anyone can test them out. Lens Studio states on its website that: "[w]ith a huge set of built-in features including custom shaders and advanced tracking technology, the possibilities are endless." And it is absolutely true. The creator has the ability to track the user's face, hands or feet. While creating 3D world lenses, the creator can decide whether the user can interact with the 3D world lens everywhere or only while the user is in a specific place or has a specific object. The creator can implement into the effects 2D images, 3D models, animations, sound or even machine learning. There is also a feature in which certain elements (animation images, etc.) can only be displayed by a specific action such as opening one's mouth or winking. This opens the door for the creation of simple games. In Lens Studio, creators do

not have to have a huge knowledge of coding for lens creation. The creators work with Script Graph Editor or Behavior helper script. For working on more sophisticated effects, Lens Studio also offers a JavaScript option (Lens Studio, 2021).

4.2 Instagram and Facebook

The social networks Instagram and Facebook both belong to the Facebook company. Both networks provide the user with a space to share their own photos, videos, and opinions, chat with friends, and much more (Facebook, 2021).

Like with Snapchat, with Facebook and Instagram, the user can take a photo directly in the application, apply an augmented reality-based effect to it and share the image with the world. The user can use the effect on both the surroundings and his face. However, compared to the effects from Snapchat, in Facebook and Instagram, we can notice a more flawed tracking of faces and objects and a more inferior quality of filters in general (Schlerf, 2020).

Although both social networks belong to one company and augmented reality effects are used in both networks for the same purpose - for sharing photos for twenty-four hours (so-called stories), the selection of filters on the Facebook platform is significantly smaller.

4.2.1 Spark AR

Instagram and Facebook started just like Snapchat - with several AR effects created by in-house developers and designers. In 2017, Facebook created the free software SparkAR studio, which is available to the general public. With this software anyone can create their own AR effects and upload them to either Facebook or Instagram. It is actually an imitation of Lens Studio by Snapchat and the software works on the same principle (Schlerf, 2020).

4.3 TikTok

The Chinese social network TikTok is one of the most used social networks of 2020 (Statista, 2021). TikTok is a social network focused primarily on sharing short videos - usually short sketches, or funny or dance videos (Herrman, 2019).

Users can upload the video directly from their gallery or create it instantly on the social network. If the user uploads the video directly on TikTok, the platform offers various functions that can diversify the video. One of these functions is AR effects working according to the same principle as those in the previous social networks. Due to the fact that TikTok is the youngest of these social networks, the quality of its AR filters also corresponds to this. The effects on TikTok are not as complex as those on Snapchat, Instagram, or Facebook, and of these platforms, TikTok has the worst tracking (Herrman, 2019).

4.3.1 Effector

TikTok is currently also working on its own AR effects software for designers and developers - Effector. The software is currently only available in Asian countries and is very limited. The latest released version only allows the creator to insert 2D images and simple animations. Unlike SparkAR, Effector offers cat face tracking by default (Virtual Creative Design, 2020).

5 Examples of AR on social networks for marketing purposes

When social networks opened the tools for creating AR effects to the general public, businesses also took advantage of this possibility. This is advantageous for brands mainly because it brings them closer to the customer. Customers interact with the brand by applying the brand's AR effect to their videos or photos and publicly sharing them. Once the user shares an image or video with the AR filter, the brand organically gets to the user's friends, who can share and interact with the AR filter too. In the almost four years of the operation of the SparkAR and Lens Studio platforms, several of the most used effects have become stabilized among the brands. Success is most celebrated by those effects that contain gamification elements - Randomizers, Quizzes and Game Filters (Gowaaa, 2021).

5.1 Color filters

The most superficial effects in creation are color filters. 3D models and complex tracking do not have to be used in them. It is a color overlay or a simple animation that gives the user's surroundings a specific color tone (Boon, 2020).

The HK International Theater Festival (on Instagram as @festivalregiony) used this type of filter for its promotion on Instagram. For their campaign Letos jinak, the organizers chose two kinds of color filters - red and green. These filters correspond to the festival's current visual. When a visitor to the festival used the filter and shared a photo on social networks, it was immediately apparent which festival the person was at.

5.2 Beauty filters

While browsing the gallery of AR effects, users are coming across a lot of beautifying filters. This type of effect should rid you of imperfections. Some filters slightly deform the face, enlarge the lips, enlarge the eyes, rid the person of their "other chin" or add makeup (McDowell, 2021).

Beautifying filters are used by cosmetic companies that use such effects to promote their products. A good example is the L'Oréal Paris brand (on Instagram as @lorealparis). The brand offers several shades of lipstick or eye shadow to try. While users try these effects, each makeup shade is supplemented with textual information about its type and color. This simplifies the process of selecting and purchasing products. Such testing of cosmetic products is also more hygienic and faster.

5.3 Game filters

Compared to the previous effects, game filters are a more popular type among the users since it is using the gamification elements like: rewards or challenges between users. There are many types of game filters. They differ from each other in terms of, for example, in-game mechanics or controls. These effects can work for one or more players at the same time. One can find game filters that work only with front or rear cameras, or that do not make this distinction at all. Game filters are often inspired by well-known concepts of typical mobile games such as Fruit Ninja, Flappy Bird, or Guitar Hero. The user can use on-screen tapping, winking, or even opening their mouth for control in such filters (Filterious, 2020).

A particularly successful game filter is "Eat On The Beat," which is connected with the Cini Minis cereal. The user uses a selfie camera. By tilting his head, he controls the character at the bottom of the screen, who tries to eat the "minis" falling from the top of the screen. The user interface of the effect shows the user's score and lives (the user can only miss three minis), and the whole effect is complemented by background music.

5.4 Randomizers

One of the gamification filters is Randomizer that uses the challenge gamification element between other users. At the end of 2019, the "Which Disney Character Are You?" filter became popular. The filter even reached the Jimmy Fallon talk show, where the host tried the filter out in front of the audience during the show. Randomizers choose for the user the best option that suits the user the best. In the case of the Disney filter, the randomizer was choosing for users one specific Disney character. In fact, the selected option is based on coincidence. Many companies have used this concept since 2020, as it is straightforward to understand, and the user can compare himself with others. Social networks are full of randomizers, and to be functional even now, they need to be supplemented with catchy graphics, animation, or a strong idea (Gowaaa, 2021).

A graphically sophisticated randomizer was created by HBO (on Instagram as @hbo) for their brand awareness. The user is to use the front camera to record himself, and when the recording starts, the filter selects which character from the HBO movies is the most similar to the user.

5.5 Quizzes

Quiz effects work according to a similar gamification principle as randomizers. In this case the user again applies the effect to their face, but this time the user interacts with the effect a little more than in the case of the randomizer. A question appears on the user's head. The user then has a limited amount of time to answer the question before the correct answer is revealed (Boon, 2020).

Aussie Theater used this type of filter for their brand awareness. In their quiz several emoticons appear above the user's head, representing the name of a musical. The user then has 10 seconds to answer which musical is represented by the emoticons.

5.6 Real World AR

If brands sell a tangible product, they can use a filter to offer the user a realistic 3D model of the product to try. In such a filter the user places the object in his environment using the rear camera. By moving their phone, users can zoom in on and away from the product. This kind of filter is usually used by furniture, home decor, or jewelry companies (Boon, 2020).

An example of this effect is the wall decoration filter of BeWooden. On the BeWooden Instagram account - @bewoodenhome, the brand offers the user the option to try three wall decorations. The user opens the filter, chooses from three types of decorations, and then by tapping and scrolling on the screen, places the decoration in their own environment.

5.7 Target tracking

Target tracking filters can be triggered upon locating a targeted image pre-determined by the brand. Targeted images can be posters, business cards, boxes or even buildings. The user is thus given exclusivity and personalization because without a targeted image or object, the user cannot interact with the filter. Target tracking can start a game or place a 3D object into the user's environment (Gowaaa, 2021).

The Oreo brand used target tracking in its limited-edition Trolls movie tie-in (on Instagram as @oreo). The user bought Oreos in the limited edition package and opened the Oreo filter on Instagram, scanned the biscuit wrapper, and a 3D character from the movie Trolls appeared on the wrapper and started dancing on the box.

5.8 Immersive experience

This type of filter places the user's face in a 3D environment or opens up a whole new world for the user when they use the rear camera. The user can thus become absolutely anything - a famous person or even food. While using the rear camera for the immersive experience, the user can, for example, get to another city through an AR portal or see a 3D fashion show at home (Boon, 2020).

This immersive experience was provided by the German company Kaufland with the "Konig" filter. In this filter the user's face is tracked and inserted into the current Kaufland advertisement. The user thus becomes part of the commercial. The filter also contains music, and various spaces from the advertisement are alternated.

6 The process of creating and publishing a randomizer in Spark AR

As I mentioned in the previous chapter, this type of filter is a big hit among brands and can be applied to almost any field. However, it is necessary to take into account the brand's target group and not underestimate the preparation for the filter to fulfill its purpose, namely organic dissemination among users. There are a lot of randomizers on social networks, so it is good to think about how to differentiate yourself. A special design or animation can help. An important part of creating a randomizer takes place in Spark AR, where all the components are put together and a filter logic is set by programming. Here is where the biggest setbacks occur, as some practices are not supported by Spark AR and must be kept in mind in order for the filter to eventually be approved and released to the public.

For this filter, I contacted the poet Honza Doležal, who has the Instagram account @zapiskyumelce, where he currently has around twenty thousand followers. He was in a situation where he was publishing his second book of poems called NEBE and wanted to use Instagram for its promotion in the most efficient and fun way possible without a large investment in paid advertising. Together we decided to create a filter for his Instagram fans to find out about the new book. The goal was for users to try the filter, have fun, and start sharing it with others. This achieves organic engagement.

6.1 Preparation

In general, for the preparation of Instagram filters, it is important to set what your initial expectations and goals are. In the case of the filter for the book NEBE, the aim was to raise awareness of the book and the poet himself. The user who opens the filter should immediately understand how it works, and who the creator is, and should be interested in sharing a video with the filter with friends. Those users who see the filter on someone's video should understand that the poet is behind the creation of the filter, and it should entice them to try the filter or visit the poet's Instagram account.

Since the poet shares mostly his new poems on his Instagram profile, we decided to do a selection of poems from the new book NEBE for the Instagram filter and entice users to buy the book and read other poems from it. We concluded that a randomizer was the best mechanism for this since it uses a simple mechanism that users already know and understand how to interact with.

An essential part of the preparation was to do research on the competition and similar filters. We conducted research on which brands use randomizers, how they work with them and what components they incorporate into the filter. We found out that most randomizers

start with a question first, followed by a roulette of options, and after a while, the roulette stops at one option that is intended for the user. In most cases, this action takes place in the head of the user. We were inspired by the well-known "Which Disney?" filter, whose mechanics we wanted to use. As for the competition, we managed to find the "Bad Poetry Day" filter, which works according to the principle of a randomizer and generates one bad poem for the user which should describe him or her. In the Czech Republic, we found one brand that uses a randomizer for brand awareness in connection with the promotion of the book *Ester and Josefína* (on Instagram @esterajosefina). They used excerpts from their book *Terapie Sdílením* in the filter. Their "Terapie Sdílením" filter is purely a roulette without an introductory question. A user who sees this filter shared somewhere may be confused as to what the purpose behind it is.



Figure 11 – Existing randomizers on Instagram

Once we agreed on the mechanics, the selection of the poems followed. The poet decided that he wanted to include sixty-two poems in the filter, so that the same poems would not be repeated to the user over and over again. Thus, the user could try and share the filter more than once. We agreed that we didn't want the roulette to start right away as with the "Terapie Sdílením" filter, but we believed that we need to let the user know what will follow next.

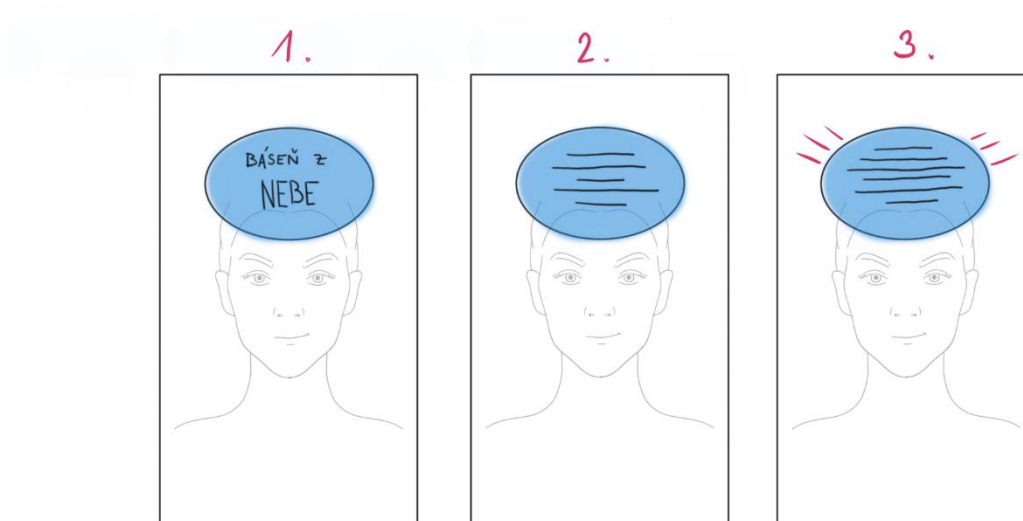


Figure 12 – Sketch of the randomizer mechanism

6.2 Design

The book NEBE was already in the process of printing when we started working on the Instagram filter. In order to maintain consistency and allow the user to connect the filter to the book, we opted for a uniform design with a book cover. The author worked on the book with the following colors: white, blue, black and red. Others were therefore out of the question. The dominant element on the cover is an analog photograph of the sky and some specific spots and lines. Most of the cover of the book consists of the huge inscription NEBE, for which the author chose a thin font. Overall, the book cover looks minimalist and the filter design should continue in this spirit.

According to the approved scheme from the preparation, it was necessary to design the appearance of individual poems, the title inscription, and the filter icon. I used Autodesk Sketchbook for the sketch designs, and once the sketches were agreed upon, I started making the designs in Figma.

I began by designing the look of the poems. The parts I worked into the design were the logo (book title), the background on which the poems would be placed, the text of the poems, and the text referring to the poet's Instagram profile. I designed 4 layouts of internal elements and 2 backgrounds.

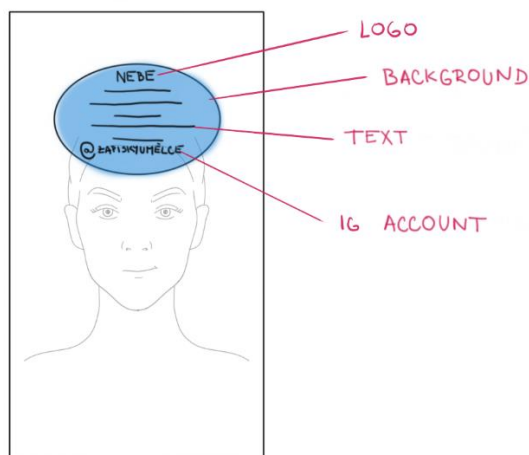


Figure 13 – Sketch of the poem's design

Layout No. 1 had the inscription "NEBE" located below the text of the poem, and the poet's Instagram account was not included in this design. Layout No. 2 had the inscription "NEBE" placed over the poem, and below the poem was a reference to the poet's Instagram account. Layout No. 3 differed from the others only in that it did not contain the inscription "NEBE". The last layout contained only the text of the poem.

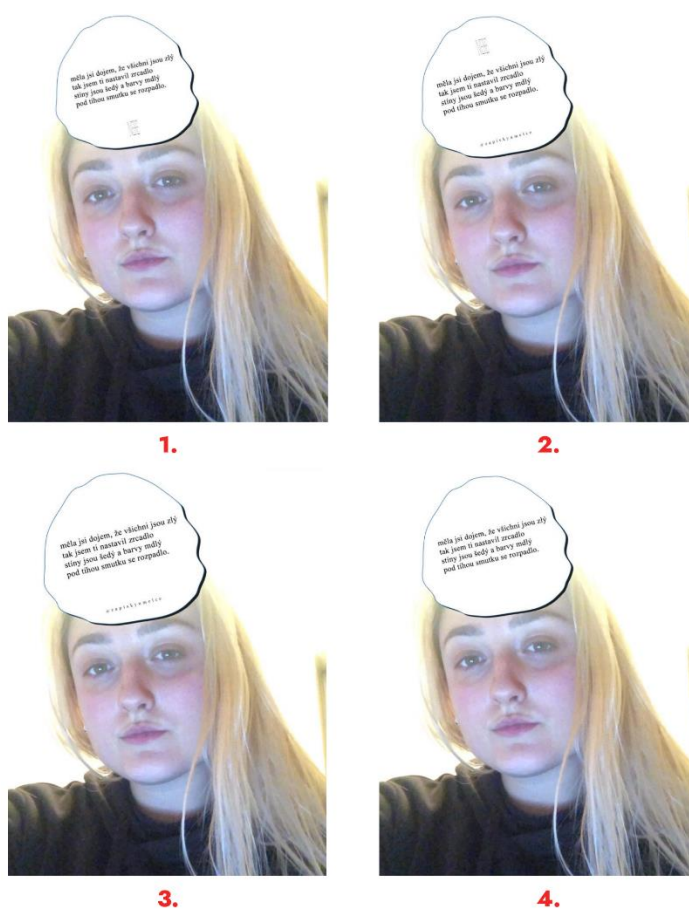


Figure 14 – Layouts of poems

Both of the background designs were based on the stain shape that was used on the cover of the book. The first design was supplemented by a blue stroke and a black line on the right side. In the second design, I worked with three shapes of stains - also from the cover of the book. I left the blue and red stains as they were on the book cover and the biggest stain, which on the cover is filled with an analog photograph of the sky, was replaced with a white background.

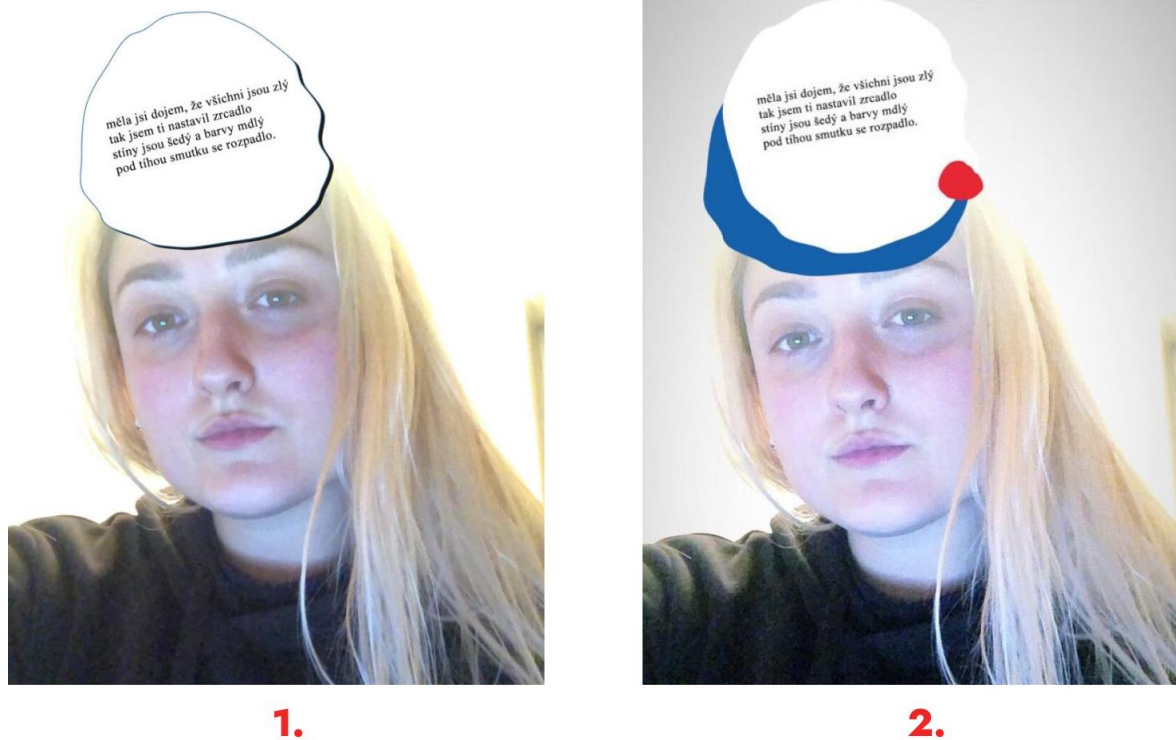


Figure 15 – Layouts of backgrounds for poems

The poet opted for a variant where the layout contained only the text of the poem. The reason was that the other elements could unnecessarily take attention away from the emotion that the poem is supposed to convey. The title of the book would be displayed in the title of the filter itself, and at the same time after clicking on the filter, each user would get to the poet's profile. Therefore, the placement of these two elements would be an unnecessary duplication.

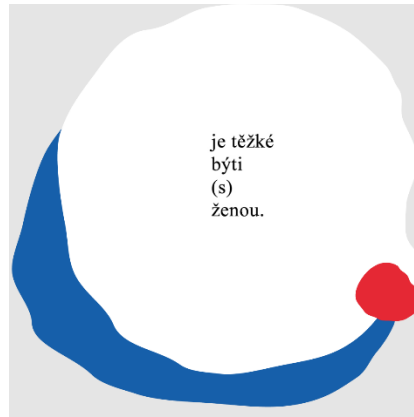


Figure 16 – Final design of poems

After creating the look of the poems, I moved on to designing the title image of the filter. From this, it should be clear what will happen in the filter and from what book the poems are. At the same time, the design was to correspond to the already prepared designs for individual poems.

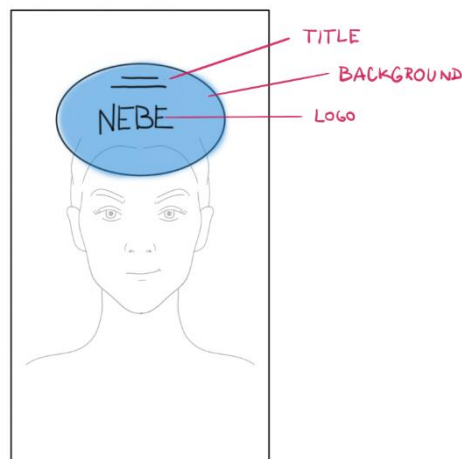


Figure 17 – Sketch of the poem's main image

Since the shape was already agreed upon based on the design of the poems, it remained the same for the title image. The only change was in the color of the main stain. For the title image, I decided to use the analog photo of the sky which is used on the book's cover. Thanks to the contrast, it will be clear to the user when the roulette starts. With this contrast, we also achieve a gradual introduction and understanding of the meaning of the filter.

For the layout in the title image, we are talking about the name of the filter and the title of the book. In the end, we managed to combine these two elements into one. We decided to use the name "Básně z NEBE" ("Poems from HEAVEN"). The name, therefore, included the "NEBE" logo. I left that in the same font as the one that was used on the book cover. For the beginning of the phrase - "Poems from" - I used a similar font but with a smaller size and I placed the text directly above the logo.

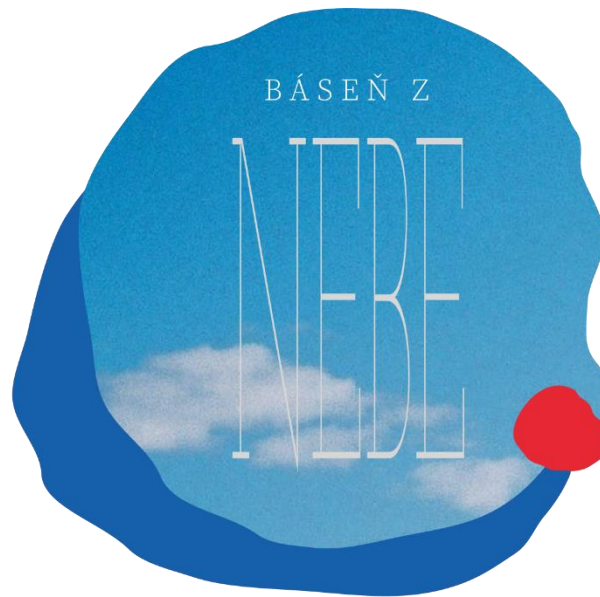


Figure 18 – Final design of the main image

To upload a filter to social networks, SparkAR requires the creation of a filter icon with certain specifications for uploading a filter to social networks, which it states on its website. The icon must be square, mustn't include transparency, and the dimensions must be at least 200 x 200 pixels. SparkAR created files for designers compatible with the Adobe Photoshop, Illustrator, and Sketch programs, and these are intended to serve as a template for creating icons. These files are available on the Spark AR website.

For the poet's filter icon, I worked with a Photoshop template. The file contains detailed layers. The layers are divided into Guidelines, Examples, and Background. The Guidelines layer needs to be turned off when exporting - it only serves the designer in the sense of letting him or her know which part will be visible on social networks. All the content that will be inside the space bounded by the green guide should be the most important and will always be visible. The space between the green and purple guides should serve as free space so that the elements in the space bounded by the green guide stand out. The red guide shows the border of the icon as it will be displayed on Instagram and Facebook. We should not place any essential element in the space between the red and purple guidelines. There are two icon designs in the Examples layer group - one with a logo and the other with an image. This group of layers must also be turned off during export because it serves only as an inspiration for the designer. The Background layer allows designers to change the background color. The icon design is imported as a new layer, and the whole file should be exported to either JPG or PNG.

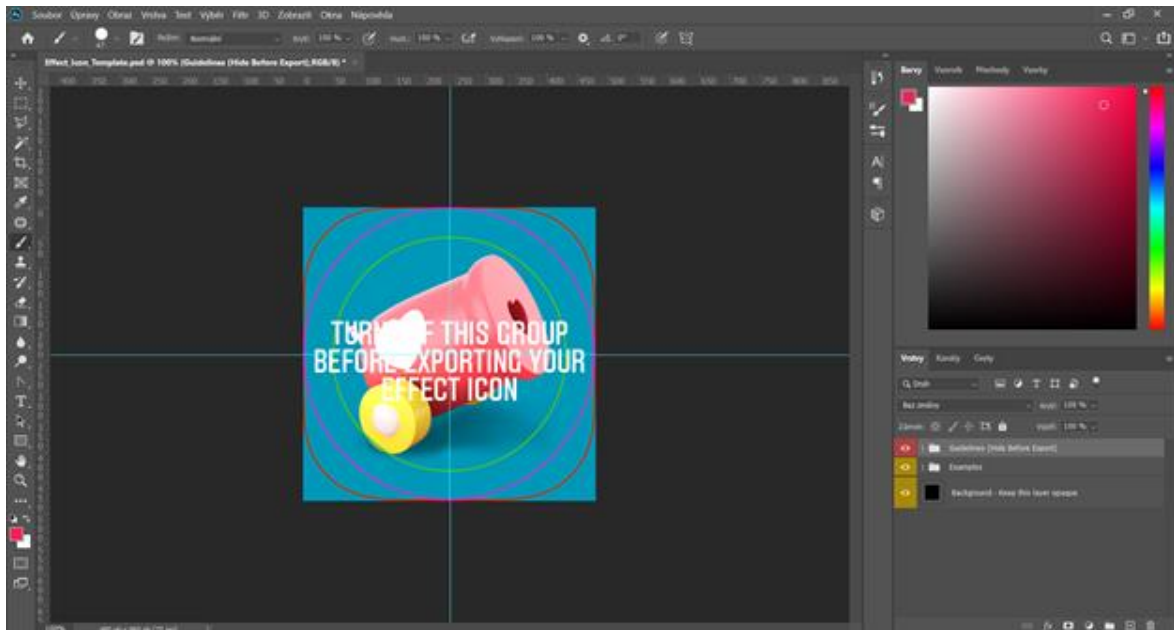


Figure 19 – Spark AR Icon template opened in Photoshop

For the filter icon, I chose the title image from the randomizer. It contains both the title of the book and the colors and elements corresponding to the book. Parts of the stain exceed the green guide, and in any case, they will be visible on social networks.

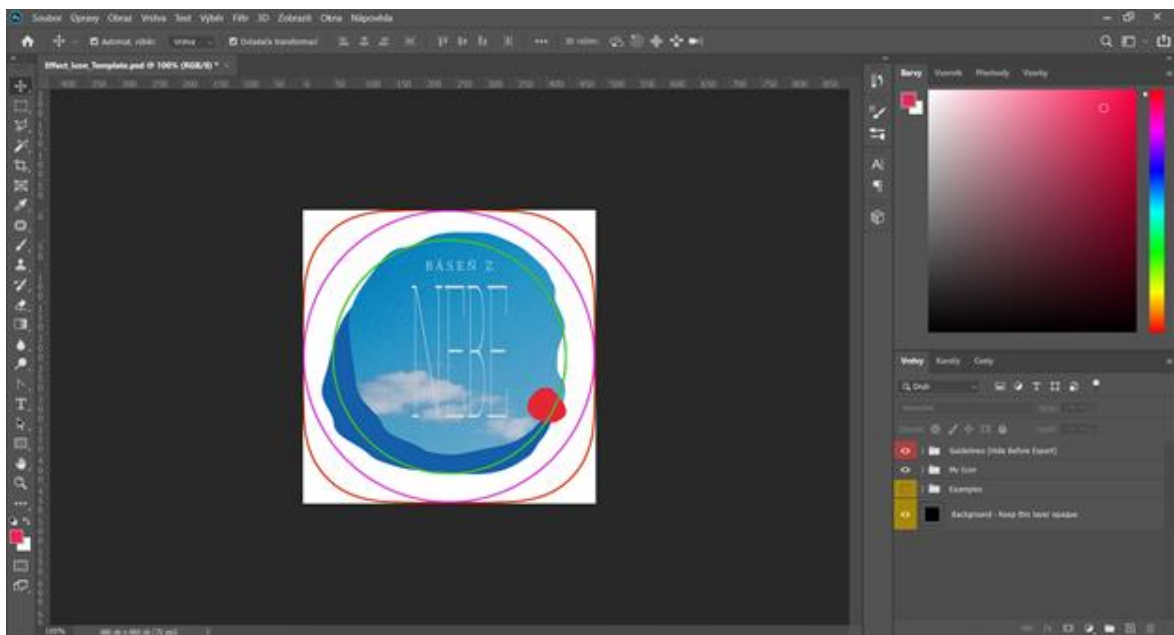


Figure 20 – Final design of the icon in Photoshop

6.3 Coding

Once I had all the graphics ready, I entered the Spark AR filter assembly phase. I downloaded the program from the official Spark AR website (sparkar.facebook.com/ar-studio/), where it is freely accessible to everyone. SparkAR identified my type of system and downloaded the corresponding version. To use SparkAR it is necessary that the computer meet these requirements:

Tab. 6.1 Requirements for Spark AR

Hardware or software component	Minimum requirement
Operating system	Windows 10 (64 bit) or MacOS 10.14+
RAM	4 GB
CPU	Intel Core i3 2.5Ghz, or AMD Bulldozer/Jaguar/Puma/Zen with SSE4.1 instruction set. M1 or newer. Processors produced in the last decade will likely support this instruction set.
GPU	Intel HD Graphics 4000/Nvidia GeForce 710/AMD Radeon HD 6450
Monitor resolution	1280 x 768

Since I work on a Windows system, the installer automatically asked me to install iTunes, which is necessary for the program to work.

After Spark AR is successfully installed, it offers thirteen templates from which to create filters - for example, Color Filter, Face Mask, Neck Decoration, and 3D Animated Poster. None of the offered templates are suitable for creating a randomizer. As soon as I want to create a new project, Spark AR asks me to choose the type of project I am about to create - Plane Tracking, Target Tracking, Face Tracking, or a Blank Project. By choosing Plane Tracking, I could place objects on a horizontal surface, like a floor or a table. A Target Tracking project anchors experiences to an image in the real world, like a poster. The Face Tracking option helps the creator with effects that respond to or augment someone's face. For this project, I chose the Blank Project option, which does not limit me with anything and does not predetermine any tools.

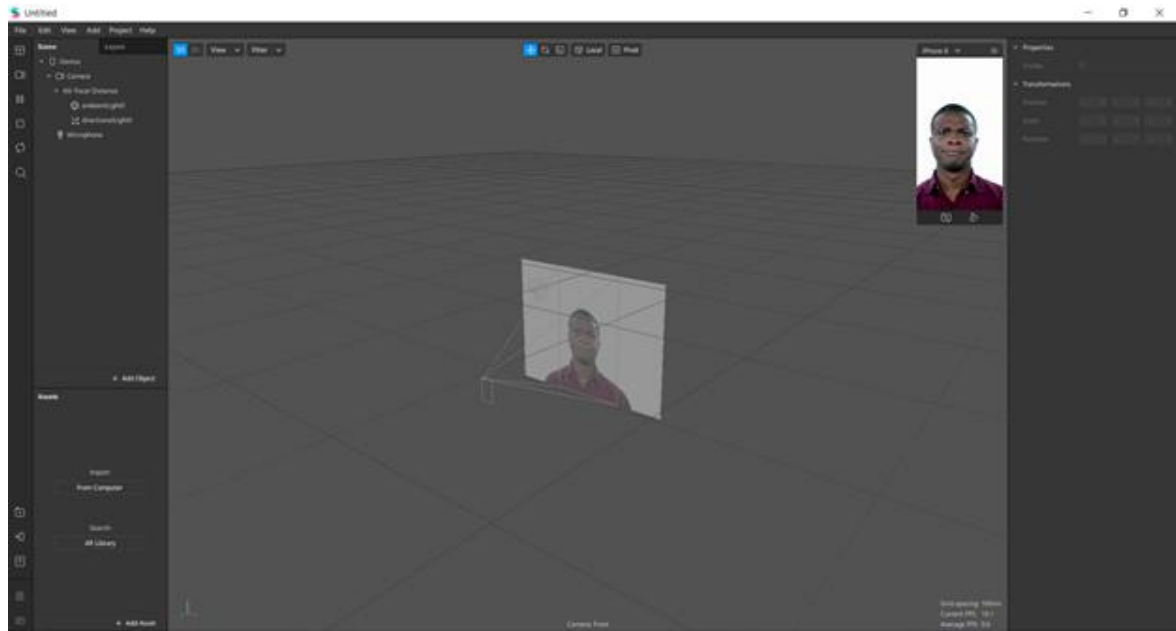


Figure 21 – Spark AR workplace

I was in a Spark AR environment. A device (meaning a mobile phone or tablet) was automatically inserted into the scene in a blank project. The device included cameras and a microphone. Cameras affect what will be displayed to the user on the screen. In the microphone section, the creator influences its functionality for the given filter. In the middle of the workspace, there is a 3D space in which all inserted assets will be placed and projected. In the upper right corner, there is a preview. For now, I kept the preview on iPhone 8, as this is the device on which I am testing.

I started by importing graphic assets that I created in Figma. These were sixty-two pictures of poems and one main picture. All the assets were pre-exported to the PNG format with transparency in a 1080x1080 px resolution with a specification of 72ppi. Since SparkAR has set a limit for effects and the size must not exceed 40MB, all assets need to be optimized to the lowest size possible. Therefore, before importing, I optimized all the images via the online image optimizer - imagecompressor.com. First, I imported the main image separately, which I inserted into the lower-left tab of the Assets and imported "From Computer". I was importing the exported poem designs into Spark Ar via Animation Sequence, which I was inserting into the Assets tab via the Add Assets button. At the time, in the Assets tab, I had the Texture group, which contained the main image, and the Animation Sequence group, which still had an empty AnimationSequence0. By clicking on the AnimationSequence0, the settings of it appeared in the upper right corner. The very first specifications were Textures. In this part, I used New Image Textures to insert all the images with poems. In the Textures group in Assets, I now had two items - the main picture and a group of all the poems. To have better control over the quality of the outputs, I clicked on each item in Textures and checked "No Compression" in the right column next to the manual compression. I know from past experience that Spark AR could compress my assets so that they are not readable at all, and this could not happen with this filter, where the dominant text was.

I had the assets imported in the program, but I had not yet applied them to the scene. Since the poet and I agreed in the design part that the images would respond to the movement of the user's head and would be placed on his forehead, I inserted a Face Tracker into the Scene tab using the Add Object button. Face Tracker is able to recognize a face and apply other objects, which the creator inserts into the scene, on the face. Under Face Tracker, I inserted a Null Object via Add Object. Then I inserted two Plane Objects and put them under the Null Object. A Null Object containing Plane Objects linked to the Face Tracker controlled the position and size of the assets that I was about to load into the plane objects on the user's face. By default, the position of the Plane Objects was in the middle of the user's face, which I did not want, and with the help of the Null Object, I changed the position X from 0 to 0.12 on the right side in the transformations column.

In order to be able to connect the prepared Textures with the Plane Objects and get them on the user's head, I had to first prepare the materials. I created them again in the tab with Assets. I created 2 materials - Start and Poems. I clicked on each material and in the options in the upper right corner I set the shader type to "flat" for both materials, as these are 2D assets. In the shader properties I set the textures. For the Start material I set the texture of the main image and for Poems I chose AnimationSequence1. The materials were ready and the connection with the Plane Objects followed. In the Scene I clicked on plane0 and in the properties I selected the material Start. For plane1 I chose the material Poems. For the first time, assets above the user's head appeared in the middle of the workspace. For better readability, I resized the NullObject in the properties. I changed the value of X and Y from 1 to 1.5.

After that the images above the head were flashing without any logic, they did not stop doing so and the first image did not even appear. Here was the start of the programming phase, which took place in the Patch Editor. The Patch Editor is located at the bottom of the workplace. I started with a Runtime patch that tracks the number of seconds that elapsed since the effect started. I created the Offset patch and linked it to the Value input of the Runtime patch. The Offset patch compares the current value and the base value and represents the offset of the current value from the base value. I continued by using the Less Than patch, where I set the duration of the whole animation to 5 seconds, and connected it with the Loop Animation patch. In Loop Animation the duration of the speed of the roulette is set. For this filter I set the duration at 0.09 seconds. I followed up on the Loop Animation patch with the Random patch, which ensures that the pictures of the poems are composed randomly. In this patch, I set the End Value to 62, as this is the number of poems that are part of the filter. From the Random patch, I continued to the Floor patch, which rounds numbers so that only one image is always displayed. In the Assets tab, I created a patch from the Animation Sequence in which all the pictures with poems are. This set the logic for the animation of the poems. I then proceeded by creating the overall logic of the filter. I started by inserting the Camera patch. I connected the video output to the Delay patch and set its duration to 2 seconds. This duration represents how many seconds the main image is going to be visible after the user presses the record button. I connected the Delay patch with the plane0 patch, which contains the main image. The last thing that had to be done was to link the animation and the initial image together. I used the Pulse patch that triggers a pulse whenever a state changes from on to off or vice versa. From the Camera patch, I led another connection from the video recording output, this time

to the Pulse patch. From the Pulse patch I led the connection to the Reset input of the already existing Offset patch. From the Pulse patch, I led another connection, this time to the Switch patch. This patch arranges for the main image to change into a sequence of poems. I created a new Delay patch and linked it to the Switch patch. I set the duration to 2 seconds. This ensured that after 2 seconds the main image is going to change into an animation of the poems. I connected the Delay patch via the Not patch to plane1, which contains the assets with poems. Finally I successfully tested the filter for the iPhone 8.

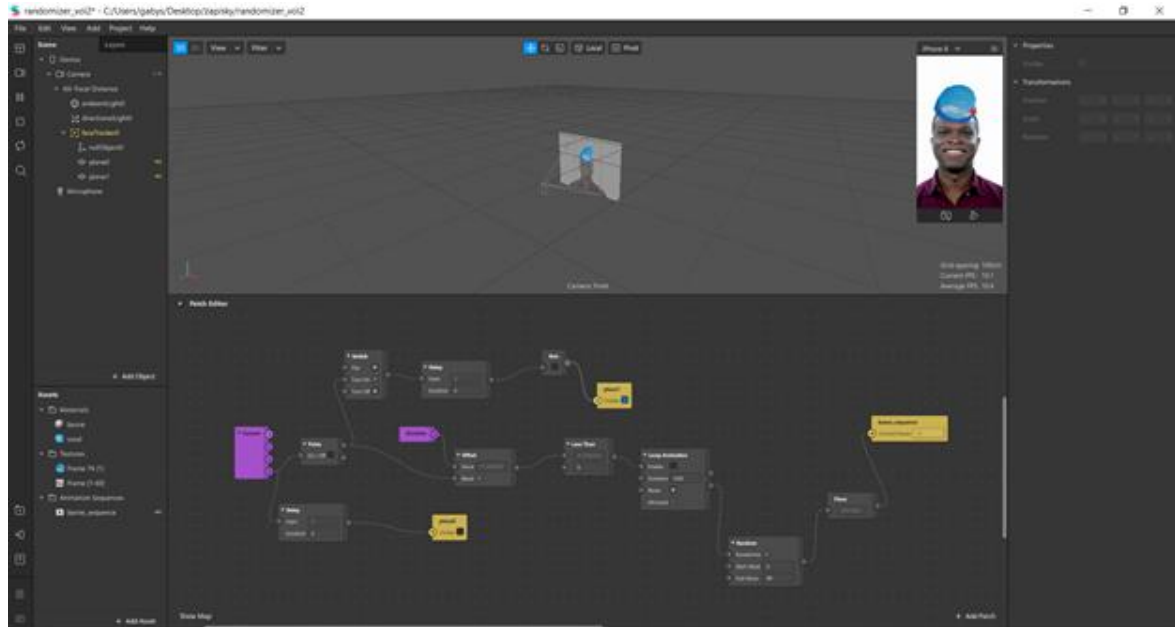


Figure 22 – Final code for the randomizer

6.4 Export and Uploading

When exporting from Spark AR the file size met all the requirements and capabilities. The filter was exported in the .arproj format. To upload a filter to social networks, it is necessary to have a preview video uploaded with the filter, which shows users how the filter works. At the same time, the video is used for gaining the approval of Facebook.

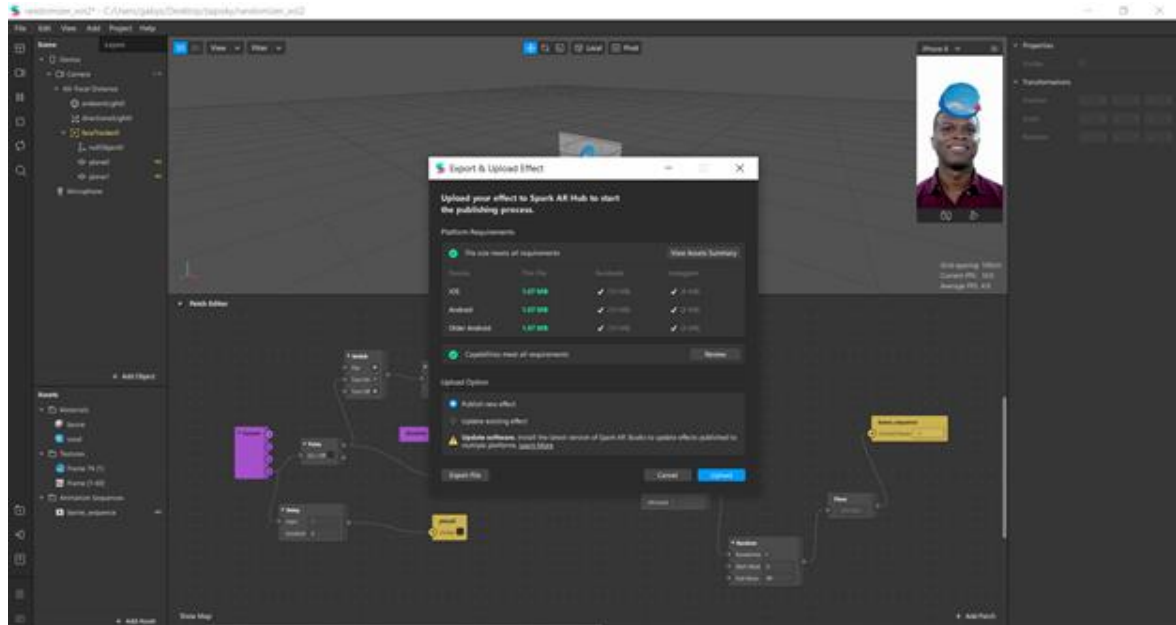


Figure 23 – Export from Spark AR

For uploading the filter to social networks, it was necessary to have access to the poet's Facebook page. After logging in to facebook.com/sparkarhub, I started filling in information about the filter and uploading files. It was necessary to enter the name of the filter and the files exported from SparkAR, choose between the Facebook and the Instagram platform - here, I chose the Instagram option, enter the filter keywords and category, and submit the demo video and the icon, which I created according to the instructions from SparkAR. After a filter is submitted, Facebook usually has 5 days to accept or reject it.

The day after the filter was uploaded, I received a notification of one issue that needed to be solved. The filter did not pass the first round of approval due to non-compliance with Spark AR policies. There were reports that the effect contained words that are shocking, disrespectful, or derogatory (Policies 3.1, 3.2, and 3.3). There were specifically two poems that included two obscenities. It was necessary to remove these two poems from the filter. This meant changes in Spark AR - concretely removing the entire material with poems and textures with poems, uploading a new group of textures (this time without the two poems), creating a new material, attaching a new group of textures to it, and reconnecting the material to plane1 under the Null Object, which falls under the Face Tracker in the scene. I had to change the value from 62 to 60 in the Random patch, and once I had adjusted all the details, I had to re-export the filter and repeat the whole upload process.

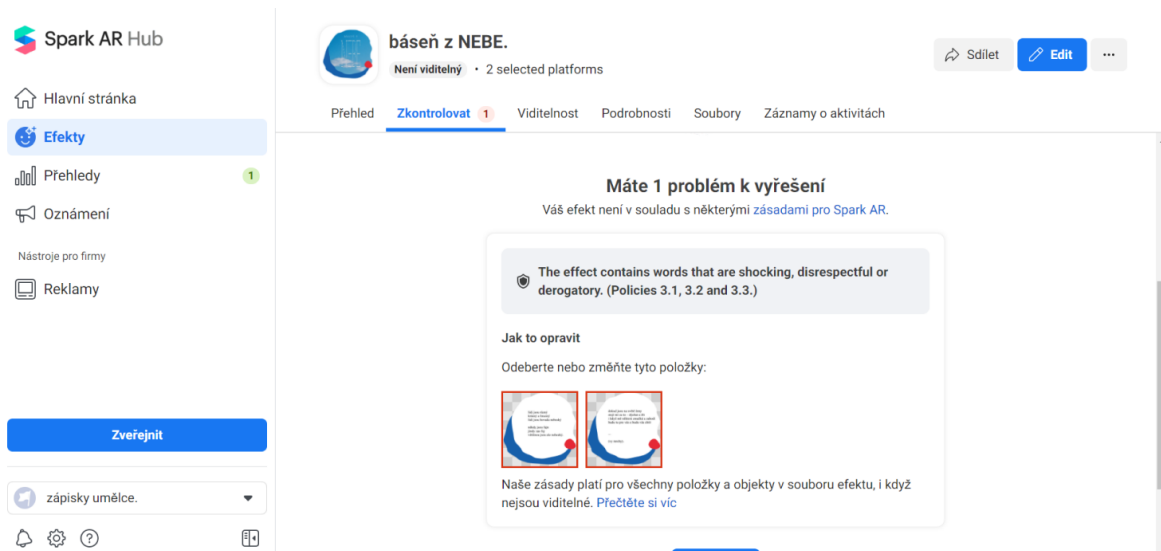


Figure 24 – Problem with Spark AR policies (vulgarism)

The next day another alert came from Spark AR. The icon did not go through the approval process this time. The icon contained the name of the effect, which again was not in accordance with the principles of Spark AR - specifically with Policy 2.2. We decided to use only a shape without text for the icon. So there are 3 stains left - red, blue, and the main one with the analog photo of the sky. In this case, all I had to do was replace the icon. There was no need to go through the whole uploading process again.

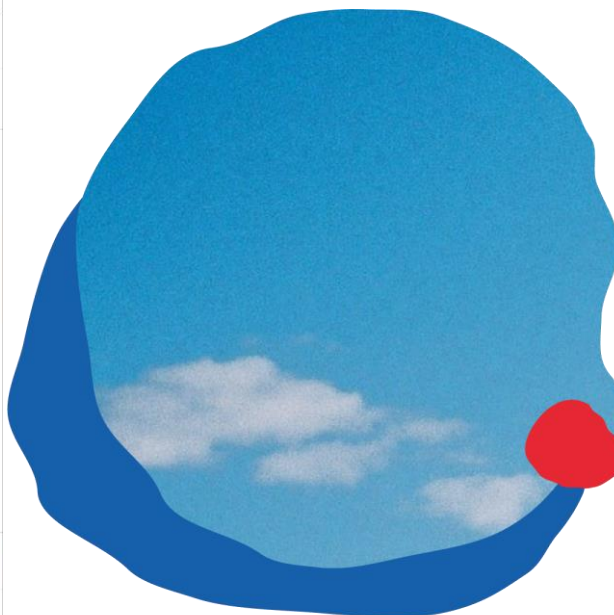
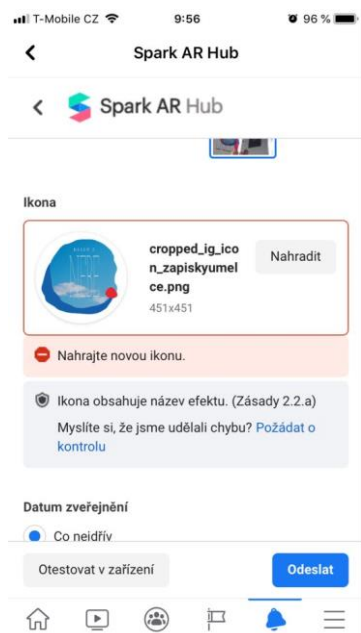


Figure 25 – Problem with Spark AR policies (icon)

Figure 26 – New icon for the filter

Five days after the last update, Spark AR announced that the effect did not pass the approval process because it promotes alcohol, tobacco, marijuana, or pharmaceutical, medical products or was published by a brand associated with such products. Specifically, there was a breach of Policy 3.3.c. At this point, Spark AR offered me either the option to edit the filter or a manual review if I thought they had made a mistake in the approval process. I decided to ask for the manual review and stated in the comment that the filter was an artistic expression in the form of poems and not a promotion of any of the mentioned products.

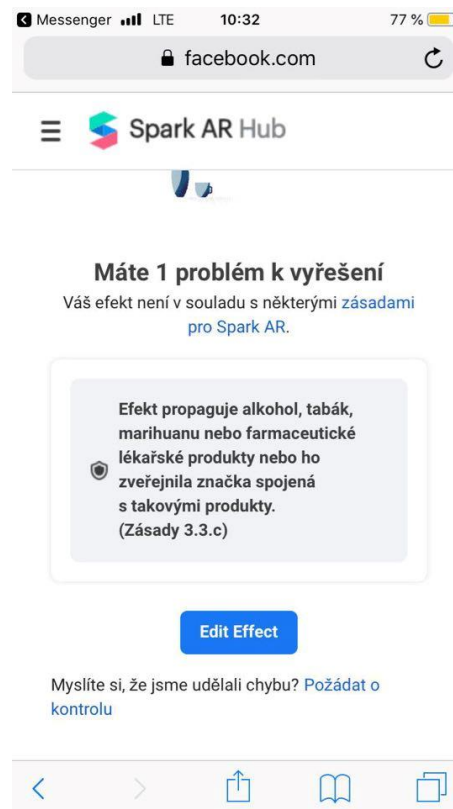


Figure 27 – Problem with Spark AR policies (alcohol and drugs)

The next day, a notification came saying that the filter had been approved. In total, the approval process took 8 days and two adjustments had to be made.



Figure 28 – Notification of acceptance

6.5 Analytics

The poet recorded himself with the filter on and published the video on his Instagram to get the filter among his followers. On the first day, the filter was opened 3300 times, 2200 videos were taken, 229 users saved the filter, and 288 users shared a video with the filter on their Instagram.

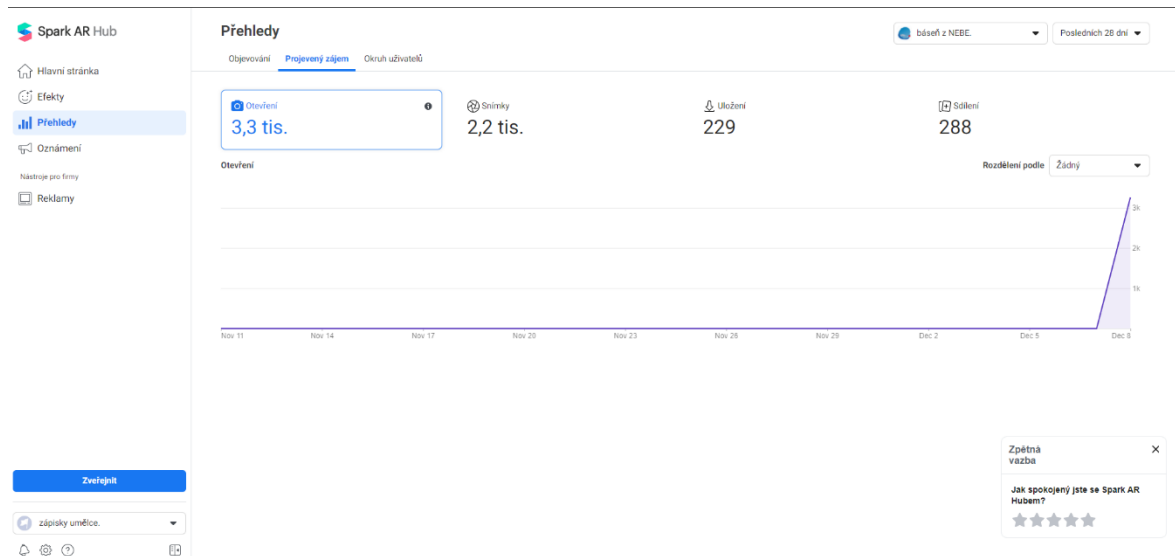


Figure 29 – Analytics day after the publication

A month after its publication, the filter achieved 20,400 impressions, 17,100 openings, and 6,800 captures, of which 1,100 videos were published publicly. The biggest peak of the filter was the first day of its publication. Users who used the filter and marked the poet's account with the poet reshared it, which could also serve as another motivation for sharing the filter.

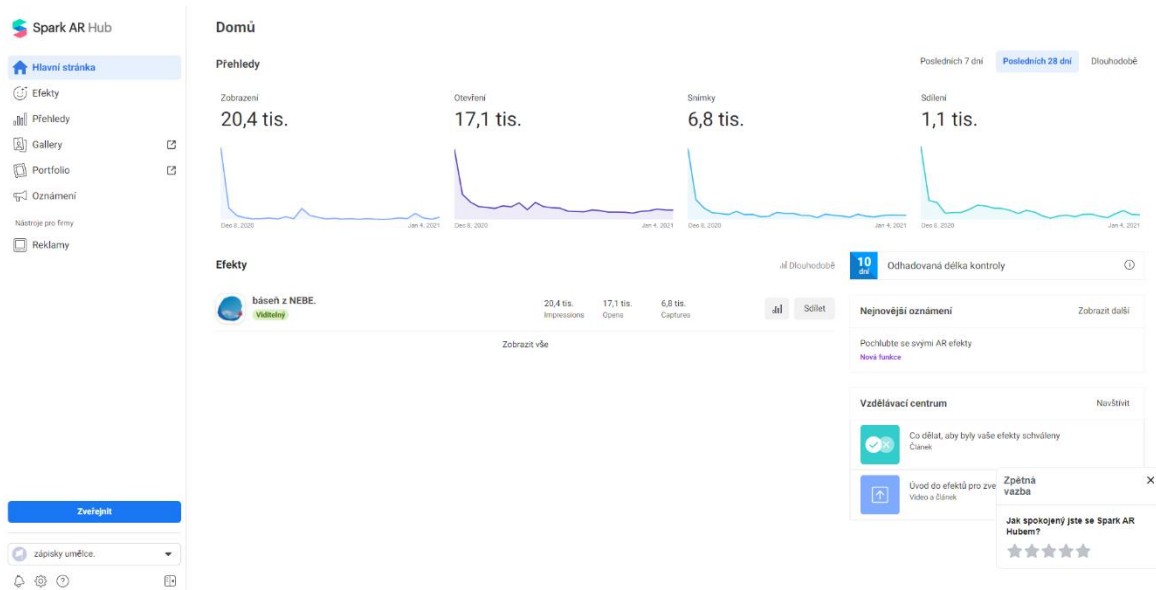


Figure 30 – Analytics after one month

It is currently half a year since its publication. According to statistics from the Spark AR Hub, the filter was viewed 34,300 times and opened 44,600 times, 14,800 videos were taken with the filter, and 2,100 videos with the filter were shared publicly on Instagram. Spark AR also offers statistics on the range of users who have used the filter. From this, it can be deduced that 3,499 users actually opened the filter - the sum of users according to gender. Because the number of openings is more than ten times higher, users were returning to the filter. At the same time, if each user shared exactly one video, it would mean that 60% of the users who opened the filter had their interest sparked so much that they decided to share it with others. The filter was mostly shared by the age group 13 - 24 years, primarily women. In addition to the Czech Republic, the filter also affected users from Slovakia and a few users from Germany.

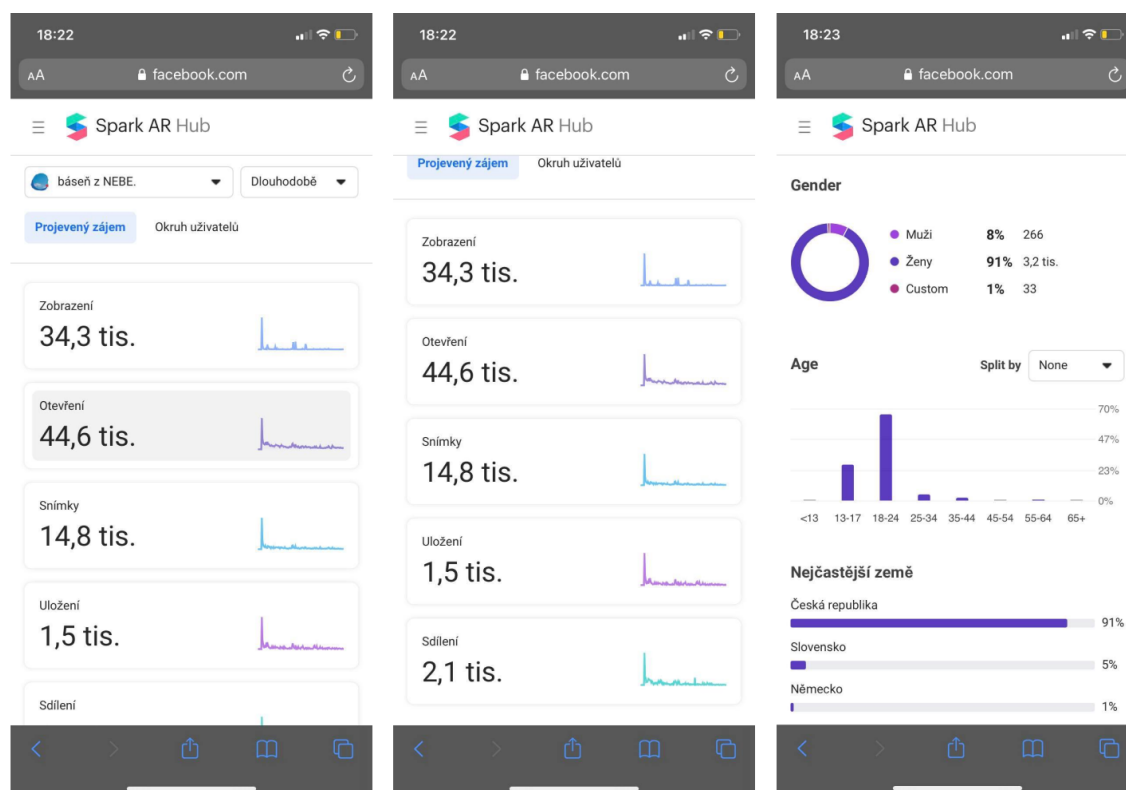


Figure 31 – Analytics after six months

These results are a real success given the poet's Instagram follower base. The poet currently has 22,500 followers, which means that more than 10% of his fans have recorded themselves with the filter and shared it. Of course, the filter could be tried by anyone who is not a fan of the poet, and therefore it is only a theoretical statement. In any case, if everyone who shared the filter had at least 100 followers, it would mean that the filter and thus the promotion of the book organically reached 210,000 users.

Conclusion

This thesis dealt with the use of augmented reality for brand promotion on social networks to achieve organic engagement. First, the history, current possibilities and practical use of augmented reality are mapped out for the purpose of setting augmented reality in context. Then the thesis dealt in more detail with the use of augmented reality on social networks, and specific examples of the use of augmented reality for promotional purposes were given. In the last part, the thesis was focused on the process of creating a promotional AR filter for social networks and then its publication and evaluation.

Augmented reality is still a substantially new technology that still has further development ahead of it, and its potential will grow not only for marketing purposes. Currently, augmented reality is primarily used for businesses and to facilitate work; however, large social networks are trying to spread it to the general public by using their augmented reality softwares. The AR filter which was created for this thesis using Spark AR by Facebook, confirmed that if a thorough creation process is followed, a high organic engagement with an AR filter can be achieved among the general public on Instagram. The success of the AR filter can also be assessed by the fact that it has been used by some users more than once.

An important finding that follows from this thesis is that it is necessary to take into account that there is a sufficient amount of time for the task when publishing AR filters on third-party platforms - namely social networks. In this work, it was clearly confirmed that the approval process may be longer than the third party states, and due to the large number of filters that are waiting for approval, some filters may be unjustifiably rejected, as was the case with the AR filter for the poet.

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